Supplementary Material for:

Forty Years of Interatomic Potentials, One Error Geometry: Hyper-Ribbon Universality and Why Pooled Benchmarks Mislead

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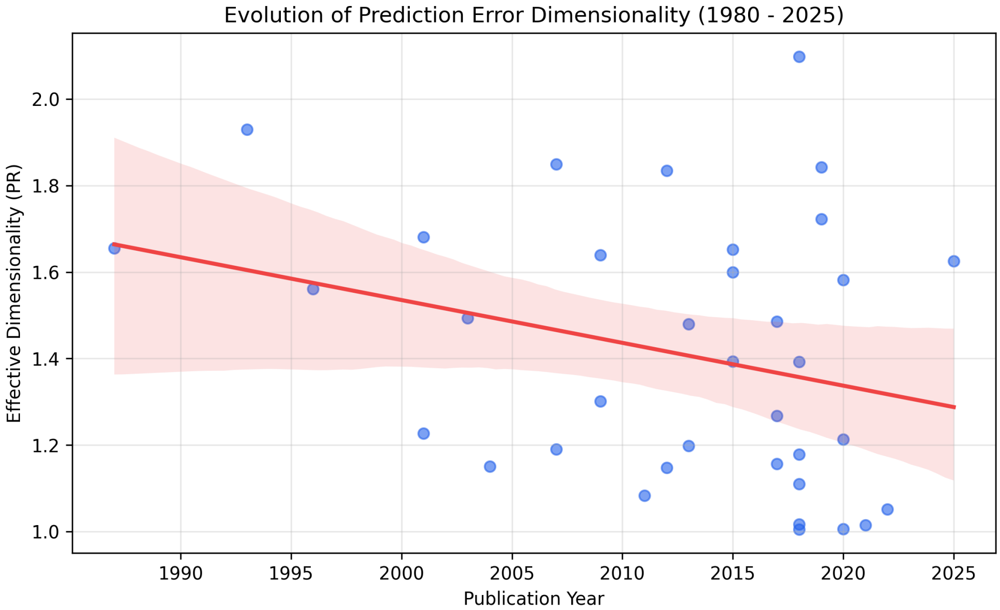
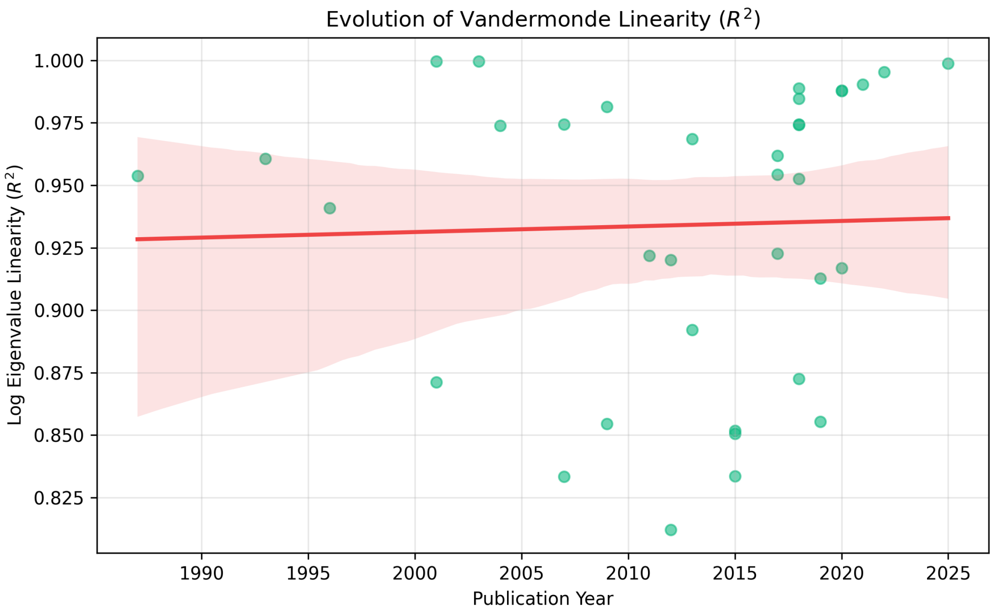
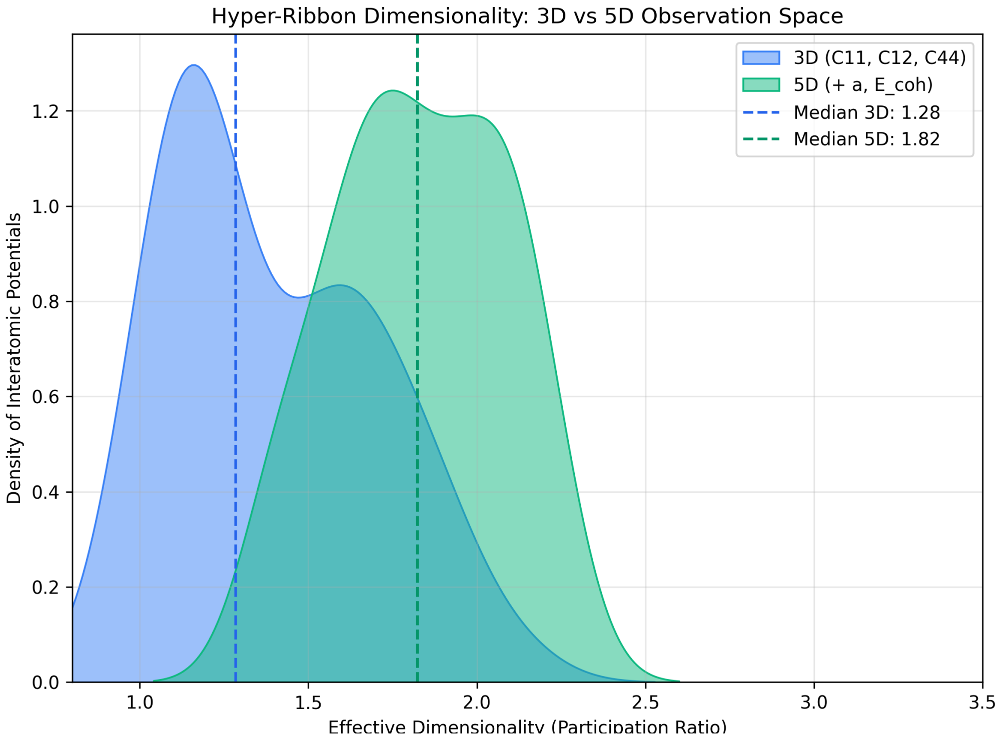
This document contains supplementary figures referenced in the main article. The main manuscript presents six figures (the journal limit); the material below provides additional analyses that support the central claims without overloading the primary narrative.



Supplementary Figure 1: Effective dimensionality (Participation Ratio) of prediction errors stratified by potential publication year (1980–2025). The hyper-ribbon structure remains stable across all decades, with PR bounded well below the ambient dimension 3 (median ≈ 1.1) regardless of model complexity or publication era.



Supplementary Figure 2: Log-eigenvalue linearity (R^2) stratified by publication year. The Vandermonde universality class signature ($R^2 > 0.80$) persists across all decades of interatomic potential development.



Supplementary Figure 3: Hyper-ribbon dimensionality distribution comparing the standard 3D elastic constant space (C_{11}, C_{12}, C_{44}) to an expanded 5D space incorporating lattice constant (a) and cohesive energy (E_{coh}). The median effective dimensionality remains highly compressed despite a 66% increase in the number of target observables, confirming that the low-dimensional error structure is fundamental to the observable space rather than an artifact of dimensionality restriction.